# 1. Systematic approach to the comatose overdose

### Resuscitate

- ABC, intubate if GCS ≤ 8 or failing airway; glucose; thiamine
- ECG early (QRS, QTc, R in aVR, bradyarrhythmia), temperature, pupils
- Naloxone trial if opioid suspected

### Assess

- History (collateral, packets, pill bottles), examination for toxidrome
- Bloods: paracetamol and salicylate levels in all, ABG, lactate, osmolar gap, CK, U&E, glucose
- Contact poison centre (TOXBASE)

### Then

- Decontamination, enhanced elimination, antidotes, organ support, psychiatric review

**Key point:** Resuscitate → ECG → levels → decontaminate → antidote → support.

# 2. Gastric decontamination

## Activated charcoal

- 50 g within 1 h of a toxic dose of an adsorbable drug; later for sustained-release or anticholinergic
- Only with protected airway; contraindicated: hydrocarbons, corrosives, ileus
- Not for metals, lithium, iron, alcohols

## Gastric lavage

- Not routine (AACT/EAPCCT 2013): aspiration, perforation, little benefit
- Rare: life-threatening amount < 1 h with no antidote (e.g. colchicine)

## Whole bowel irrigation

- Body packers, sustained-release, iron, lithium: PEG 1–2 L/h

**Key point:** Charcoal yes if early and airway safe; lavage almost never.

# 3. Toxidromes

## Patterns

- **Opioid**: coma, miosis, ↓ RR
- **Cholinergic** (organophosphates): SLUDGE/DUMBELS, bradycardia, miosis, bronchorrhoea
- **Anticholinergic** (TCA, antihistamines): hot, dry, red, mydriasis, tachycardia, retention, delirium
- **Sympathomimetic** (cocaine, amphetamine): hyperthermia, sweating, tachycardia, mydriasis
- **Sedative-hypnotic**: coma, normal pupils, normal vitals
- **Serotonin**: clonus, hyperreflexia, hyperthermia

## Brady + hypotension + hypothermia

- β-blocker, CCB, digoxin, clonidine, opioids, sedatives, organophosphate, hypothyroid

**Key point:** Pupils, skin, temperature and heart rate sort most toxidromes.

# 4. β-blocker and calcium channel blocker

## Features

- Bradycardia, hypotension, AV block; CCB → hyperglycaemia (insulin release blocked); β-blocker → hypoglycaemia; propranolol → seizures, wide QRS

## Treatment

- Atropine, fluids, calcium (CaCl 10% 10–20 mL, repeat), glucagon 5–10 mg (β-blocker)
- **High-dose insulin euglycaemia** 1 U/kg bolus then 1–10 U/kg/h + dextrose, K monitoring
- Vasopressors/inotropes (adrenaline), pacing
- Lipid emulsion 20% 1.5 mL/kg if refractory (lipophilic drug)
- VA-ECMO for refractory shock; methylene blue for vasoplegia

**Key point:** Calcium, high-dose insulin, catecholamines, then ECMO.

# 5. Digoxin

### Features

- Nausea, visual (yellow) changes, any arrhythmia (bidirectional VT, AV block with atrial tachycardia), hyperkalaemia in acute toxicity

### Digoxin-specific Fab

- Indications: life-threatening arrhythmia, K > 5–6 (acute), level > 10–15 ng/mL (acute) or > 6 (chronic), end-organ damage
- Dose (vials) = level (ng/mL) × weight (kg) / 100; or 10 vials empirically in acute

### Avoid

- Calcium in hyperkalaemia (traditional caution), cardioversion if possible; correct Mg and K

**Key point:** Hyperkalaemia in acute digoxin poisoning = give Fab.

# 6. Tricyclic antidepressants

## Features

- Na channel block: QRS > 100 ms (seizures), > 160 ms (VT), R in aVR > 3 mm
- Anticholinergic, α-block hypotension, seizures, coma

## Treatment

- **Sodium bicarbonate** 1–2 mmol/kg boluses to pH 7.45–7.55 and narrow QRS
- Hyperventilate if intubated; benzodiazepines for seizures (no phenytoin)
- Noradrenaline for hypotension; lidocaine for VT; lipid emulsion if arrest
- Avoid class Ia/Ic, flumazenil, physostigmine

**Key point:** TCA: bicarbonate and benzodiazepines.

# 7. Enhanced elimination and others

### Haemodialysis (EXTRIP)

- Lithium, salicylate, toxic alcohols, metformin (MALA), valproate, theophylline, barbiturates

### Urinary alkalinisation

- Salicylate (urine pH 7.5–8.5); phenobarbital

### Specific antidotes

- Paracetamol → NAC; opioids → naloxone; benzodiazepines → flumazenil (rarely); organophosphates → atropine + pralidoxime; cyanide → hydroxocobalamin; CO → O2 / hyperbaric; methaemoglobin → methylene blue

**Key point:** Know which poisons are dialysable: "I STUMBLE" / EXTRIP list.